

RTX 3090 Local Inference Cost Analysis

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Executive Summary

This document provides a detailed, empirically-verified cost analysis for running local LLM inference on an NVIDIA RTX 3090 GPU via llama.cpp, compared to cloud API pricing. All figures are based on **measured benchmarks** from the production server, not estimates.

Bottom line: Local inference costs **\$0.04/M input tokens** and **\$0.15/M output tokens** — a **99.7–99.8% savings** over Claude Opus 4.6 cloud pricing.

1. Hardware Configuration

Component	Detail
GPU	NVIDIA RTX 3090 (24GB VRAM)
Acquisition Cost	~\$900 (purchased April 2025)
TDP	350W (spec), ~300W sustained inference load
Host	o11a (VM 101 on Proxmox)
Software	llama.cpp (llama-server)
Model	Qwen 3.5 (GGUF quantized)
Context Window	262,144 tokens
Location	Louisiana, USA

2. Measured Performance

Benchmarks were taken directly from the llama-server `/v1/chat/completions` endpoint with timing data on April 5, 2026.

Prefill (Input/Prompt Processing)

- **Speed:** 493.7 tokens/second
- **Workload type:** Compute-bound, parallel batch processing
- **What it does:** Processes the entire input prompt through the model in one pass

Decode (Output/Token Generation)

- **Speed:** 122.5 tokens/second
- **Workload type:** Memory-bandwidth-bound, sequential (one token at a time)
- **What it does:** Generates response tokens autoregressively

Why They Differ

Prefill processes all input tokens simultaneously using the GPU's parallel compute cores (tensor cores, CUDA cores). Decode generates tokens one at a time, bottlenecked by memory bandwidth as it reads the full KV cache for each token. This is why prefill is ~4x faster than decode on the 3090.

3. Electricity Cost

Parameter	Value
Electricity Rate	\$0.155/kWh (Louisiana residential, verified)
GPU Power Draw (inference)	~300W sustained (measured average under load)
GPU Power Draw (idle)	~15-25W

Cost Per Million Input Tokens (Prefill)

Time = 1,000,000 tokens $\div$ 493.7 tok/s = 2,025.5 seconds = 0.5627 hours
Energy = 0.300 kW $\times$ 0.5627 hours = 0.1688 kWh
Cost = 0.1688 kWh $\times$ \$0.155/kWh = \$0.0262

Electricity cost per 1M input tokens: \$0.026

Cost Per Million Output Tokens (Decode)

Time = 1,000,000 tokens $\div$ 122.5 tok/s = 8,163.3 seconds = 2.2676 hours
Energy = 0.300 kW $\times$ 2.2676 hours = 0.6803 kWh
Cost = 0.6803 kWh $\times$ \$0.155/kWh = \$0.1054

Electricity cost per 1M output tokens: \$0.105

4. Hardware Depreciation

Using straight-line depreciation over 5 years (20% annual rate):

Annual depreciation = \$900 $\times$ 20% = \$180/year

Hourly depreciation = \$180 $\div$ 8,760 hours/year = \$0.02055/hour

Depreciation Per Million Input Tokens

Time = 0.5627 hours

Cost = 0.5627 $\times$ \$0.02055 = \$0.0116

Depreciation per 1M input tokens: \$0.012

Depreciation Per Million Output Tokens

Time = 2.2676 hours

Cost = 2.2676 × \$0.02055 = \$0.0466

Depreciation per 1M output tokens: **\$0.047**

5. Total Cost Per Million Tokens

Cost Component	Input (per 1M)	Output (per 1M)
Electricity	\$0.026	\$0.105
Hardware Depreciation	\$0.012	\$0.047
Total	\$0.038	\$0.152
Rounded (for config)	\$0.04	\$0.15

6. Cloud Comparison

vs. Claude Opus 4.6 (Anthropic)

Metric	Local (3090)	Cloud	Savings
Input per 1M tokens	\$0.04	\$15.00	99.73%
Output per 1M tokens	\$0.15	\$75.00	99.80%
Blended (50/50 I/O)	\$0.095	\$45.00	99.79%

vs. GPT-5-mini (OpenAI)

Metric	Local (3090)	Cloud	Savings
Input per 1M tokens	\$0.04	\$0.30	86.7%
Output per 1M tokens	\$0.15	\$1.25	88.0%

vs. GPT-5.2 Codex (OpenAI)

Metric	Local (3090)	Cloud	Savings
Input per 1M tokens	\$0.04	\$2.50	98.4%
Output per 1M tokens	\$0.15	\$10.00	98.5%

Monthly Projection (estimated 75M tokens/month, 60% input / 40% output)

Provider	Input Cost	Output Cost	Total Monthly
Local (3090)	\$1.80	\$4.56	\$6.36

Provider	Input Cost	Output Cost	Total Monthly
Claude Opus 4.6	\$675.00	\$2,250.00	\$2,925.00
GPT-5.2 Codex	\$112.50	\$300.00	\$412.50

Annual savings vs Claude Opus 4.6: ~\$35,024 Annual savings vs GPT-5.2 Codex: ~\$4,874

7. OpenClaw Configuration

The cost values are configured in `openclaw.json` under the model definition. These values are in **dollars per million tokens**:

```
{
  "id": "llamacpp",
  "name": "llamacpp (local)",
  "cost": {
    "input": 0.04,
    "output": 0.15,
    "cacheRead": 0,
    "cacheWrite": 0
  }
}
```

Path: `models.providers.olla.models[0].cost`

What These Values Do

- OpenClaw uses these to track and report usage costs in `/status` and cost reports
- The daily cost report cron job references these rates
- Monthly invoices are generated using these figures
- `cacheRead` and `cacheWrite` are \$0 because KV cache operations are local with no additional cost

8. Assumptions & Caveats

1. **Power draw is estimated at 300W** — actual draw varies by batch size, context length, and quantization. True measurement requires a power meter (Kill-A-Watt or similar).
2. **Depreciation is simplified** — 20% straight-line over 5 years. GPU resale value may differ. A \$900 GPU in 2025 might retain \$400-500 value after 2 years, or could be worth less if newer generations arrive.
3. **Idle power not included** — If the server runs 24/7 regardless of inference load, idle power (~20W) is a fixed cost not attributable to per-token pricing. At \$0.155/kWh, idle costs ~\$27/year.

4. **No cooling/overhead** — Doesn't include AC costs for heat dissipation, network power, or host system power draw (CPU, RAM, PSU efficiency loss).
 5. **Model-dependent** — These numbers are specific to Qwen 3.5 GGUF on this hardware. Larger models (70B+) would be slower and cost more per token. Smaller models would be faster and cheaper.
 6. **Cloud comparison uses list pricing** — Volume discounts, committed use, or batch API pricing would narrow the gap somewhat.
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9. Recommendations

1. **Install a Kill-A-Watt meter** on the GPU server to measure actual wall power during inference — would make the electricity calculation exact.
 2. **Enable --metrics flag** on llama-server to get continuous throughput monitoring without manual benchmarks.
 3. **Consider idle cost** — If the server is dedicated to inference, factor in 24/7 idle power as a fixed monthly cost (~\$2.25/month).
 4. **Review quarterly** — Electricity rates, hardware value, and model performance change. Recalculate every 3 months.
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Appendix: Benchmark Raw Data

Test Date: April 5, 2026, 23:30 UTC **Server:** olla:11434 (llama-server) **Model:** qwen3.5:latest.gguf

Run 1 (Cold)

Prompt tokens: 28 | Completion tokens: 12
Prefill: 160.8 tok/s (174ms)
Decode: 107.6 tok/s (112ms)

Run 2 (Warm)

Prompt tokens: 30 | Completion tokens: 271
Prefill: 493.7 tok/s (61ms)
Decode: 122.5 tok/s (2,211ms)

Used warm-run values for cost calculations (representative of typical workload with warm KV cache).

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